

# Cedar Site Bai

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## ★ SUMMARY

Ph.D. cand. in CS with a strong background in machine learning, optimization, and statistics. Experienced in their applications to language model pre-training, fine-tuning, recommendation, and AI safety, with **oral presentations at ICML, ICLR, COLT** and **internships at Amazon, TikTok**. Seeking a **Researcher/Scientist** role.

## 🎓 EDUCATION

- **Purdue University** Jan. 2021 – May. 2026 (Exp.)  
*Ph.D. in Computer Science (Machine Learning & Optimization)* Advisor: [Prof. Brian Bullins](#)  
*M.S. in Statistics & Computer Science (along the way)* West Lafayette, IN, US
- **Xi'an Jiaotong University (XJTU)** Sep. 2016 – Jun. 2020  
*B.S. in Computer Science (Qian Xuesen Honors College)* Xi'an, China  
Selected Awards: "Siyuan" Scholarship (twice); National 2<sup>nd</sup> Prize in Mathematical Contest in Modeling (CUMCM, top 3%)

## 🧰 INDUSTRY EXPERIENCE

- **Amazon**, Prime Video Personalization, Applied Scientist Intern May. 2025 – Sep. 2025  
Work on **conversational recommendation** Mentors: Drs. [Duanshun Li](#), [Zhenyu Liao](#)
  - Designed a reward quantifying information gain via uncertainty reduction in multi-turn conversations
  - Built a data-generation pipeline integrating LLM recommenders, judges, and user simulators using real-world data
  - Fine-tuned LLMs with SFT, DPO, PPO, increasing conversation efficiency by 7% and hit rate by 2%
- **TikTok**, Shop Mall Ads, Machine Learning Engineer Intern May. 2024 – Aug. 2024  
Work on **ads recommendation ranking models** Mentor: [Du Zhang](#), Manager: [Dr. Yan Yan](#)
  - Delivered two launches that improved advertiser value by 1.63% and 5.94% (**\$X00,000 in profits**) after A/B test
  - Applied neural architecture search to select features/emb. sizes, reducing the model size & improving accuracy
  - Adapted Deep Cross Network for improved conversion rate (CVR) prediction on sequential data

## 📖 SELECTED PUBLICATION

- [1] **[ICML'25, Oral (Top 1%)] C. S. Bai\***, A. Y. Zheng\*, B. Bullins, R. A. Yeh. Model Immunization from a Condition Number Perspective. [\[PDF\]](#)
- [2] **[ICML'25] C. S. Bai\***, X. Luo\*, B. Li\*, P. Drineas, R. Zhang, B. Bullins. Stacey: Promoting Stochastic Steepest Descent via Accelerated  $\ell_p$ -Smooth Nonconvex Optimization. [\[PDF\]](#)
- [3] **[COLT'25] C. S. Bai**, B. Bullins. Faster Acceleration for Steepest Descent. [\[PDF\]](#)
- [4] **[ICLR'25, Oral (Top 1.7%)] C. S. Bai**, B. Bullins. Tight Lower Bounds under Asymmetric High-Order Hölder Smoothness and Uniform Convexity. [\[PDF\]](#)
- [5] **[IJCAI'21] H. Zhang, S. Bai**, X. Lan, D. Hsu, N. Zheng. Hindsight Trust Region Policy Optimization. [\[PDF\]](#)

## 🔧 PROFESSIONAL TECHNIQUES

**Machine learning (ML):** Optimizer design/analysis, reinforcement learning/RLHF, recommendation, unlearning;  
**Programming:** Python, MATLAB (**Proficient**); Shell script, C/C++, R, Java, SQL, HTML;  
**Packages / Tools:** PyTorch, NumPy, Linux Command, Git (**Proficient**); AWS, scikit-learn, TensorFlow, Pandas, Slurm.

## ADDITIONAL PUBLICATION

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- [1] **[Preprint]** C. S. Bai, A. Y. Zheng, R. A. Yeh, B. Bullins. Spectral Saliency for Machine Unlearning.
- [2] **[Preprint]** Z. Song, C. S. Bai, Z. Zhang, B. Bullins, D. Gleich. Decoupling Variance and Scale-Invariant Updates in Adaptive Gradient Descent for Unified Vector and Matrix Optimization.
- [3] **[Preprint]** H. Zhang, T. Hu, C. S. Bai, A. Xiao, D. Hsu. Uncertainty Estimation for Visual Grounding in the Open World.
- [4] **[ICLR'24]** S. Bai, B. Bullins. Local Composite Saddle Point Optimization. [\[PDF\]](#)
- [5] **[TMLR'24]** S. Bai, C. Ke, J. Honorio. On the Dual Problem of Convexified Convolutional Neural Networks. [\[PDF\]](#)
- [6] **[IROS'19]** H. Zhang, X. Lan, S. Bai, X. Zhou, Z. Tian, N. Zheng. ROI-based Robotic Grasp Detection for Object Overlapping Scenes. [\[PDF\]](#)
- [7] **[IROS'19]** H. Zhang, X. Lan, S. Bai, L. Wan, C. Yang, N. Zheng. A Multi-task Convolutional Neural Network for Autonomous Robotic Grasping in Object Stacking Scenes. [\[PDF\]](#)

## SELECTED COURSES

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**Machine Learning (ML):** Statistical ML, ML Theory, Computer Vision, Data Mining, Artificial Intelligence

**Math & Stats:** Optimization, Probability, Statistical Inference, Computational Stats, Multivariate Analysis

**Computer Science:** Data Structures, Algorithms, Database, Machine Structure, Operating Systems, Networks

## TEACHING & ACADEMIC SERVICE

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Purdue CS571 Artificial Intelligence

TA (Fall 23)

Purdue CS182 Foundations of Computer Science (Discrete Mathematics)

TA (Spring 23)

Purdue CS251 Data Structures and Algorithms

TA (Fall 21, Spring 22, Fall 22)

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- ICLR 2026, 2025, 2024
  - NeurIPS 2025, 2024, 2023
  - ICML 2026, 2025, 2024
  - AISTATS 2025, 2023

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- AAAI 2026, 2025
- TPAMI
- TMLR

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